

DCF: Firm income tax Insolvency in the binomial model, I

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What happens after illiquidity at $t + 1$?

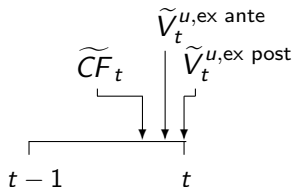
Default: Asset¹ sale.

complete sale Then the firm does not belong to its owners anymore. Only $\tilde{D}_{t+1} = 0$ useful.

partial sale a_{t+1} ² Business model not changed (point-in-time principle). Future cash flows reduced proportionally.

¹Machinery, inventories, productive assets. . .

²Naturally, $a \in (0, 1)$.



Earlier timing of model does not fit: values and cash flows are different.

We distinguish
ex ante (illiquidity occurred, no assets sold) and
ex post (assets sold, illiquidity avoided) cash flows and values.



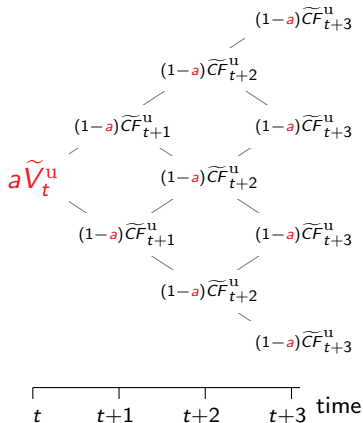
For all $s \geq 1$ we have

$$\widetilde{CF}_{t+s}^{\text{ex post}} = \begin{cases} \widetilde{CF}_{t+s}^{\text{ex ante}} & \text{no default in } t, \\ (1 - a_t) \widetilde{CF}_{t+s}^{\text{ex ante}} & \text{default in } t. \end{cases}$$

Analogously for values.

Cash flows might not be Markovian anymore.





Assumption: A sale of a assets results in a **proportional reduction** of ex ante future cash flow (=business model continues).



Ex ante cash flow generated before a possible asset sale, therefore

$$\widetilde{CF}_{t+1}^{u,\text{ex ante}}(d) = k^{E,u} \widetilde{V}_{t+1}^{u,\text{ex ante}}(d) = k^{E,u} d \widetilde{V}_t^u. \quad (1)$$

Correspondingly,

$$\widetilde{V}_{t+1}^{u,\text{ex ante}}(d) = d \widetilde{V}_t^u. \quad (2)$$



The general solution: debt satisfies fundamental theorem even in the case of default:

$$\tilde{D}_t = \frac{\overbrace{(1 + c_t) \tilde{D}_t Q(u)}^{\text{up state (liquid)}} + \overbrace{(\tilde{C}F_{t+1}^1(d) + \tilde{D}_{t+1}(d) + a_{t+1} \tilde{V}_{t+1}^1(d)) Q(d)}^{\text{down state (illiquid)}}}{1 + r_f}.$$

(Cash flows are ex post, reduced by earlier default!)

One equation, **two unknowns**: coupon c and share a of assets sold.

This single pricing equation must therefore be used conditionally.



- ① First try partial sale with $c_t = r_f$ and determine a ;
now if $a \notin (0, 1)^3$,
- ② set $a_{t+1} = 1$ and determine $c_t > r_f$.

This order is not arbitrary. We now show that the reverse order is not meaningful.

³In this case the partial sale does not suffice to close the liquidity gap.

Suppose that debt is increased until the firm becomes illiquid for the first time on the down state (this is *ex ante*)

$$\widetilde{CF}_{t+1}^1(d) - (1 + r_f)\widetilde{D}_t + \widetilde{D}_{t+1}(d) = 0.$$

Assume, we have to sell entire firm und grant no further credit *ex post*: $a_{t+1} = 1$ and $\widetilde{D}_{t+1} = 0$.

If company is *ex ante* not overindebted ($\widetilde{D}_{t+1} < \widetilde{V}_{t+1}^u$) coupon is given by equation from slide 7 as

$$c_t < r_f.$$

This is **not a feasible coupon**.



Straightforward:

$$\begin{aligned}
 0 &= (1 + c_t) \tilde{D}_t Q(u) + (\widetilde{CF}_{t+1}^1(d) + 0 + 1 \tilde{V}_{t+1}^1(d)) Q(d) - (1 + r_f) \tilde{D}_t \\
 0 &> (1 + c_t) \tilde{D}_t Q(u) + (\widetilde{CF}_{t+1}^1(d) + \tilde{D}_{t+1}(d)) Q(d) - (1 + r_f) \tilde{D}_t \\
 0 &> (c_t - r_f) \tilde{D}_t Q(u) + \underbrace{(\widetilde{CF}_{t+1}^1(d) - (1 + r_f) \tilde{D}_t + \tilde{D}_{t+1}(d))}_{=0} Q(d) \\
 0 &> c_t - r_f
 \end{aligned}$$



If company is not overindebted,

there is **exactly one of two cases**:

coupon $c = r_f$ If a share of $a < 100\%$ is sufficient to close liquidity gap, then $c = r_f$ will be the solution.

share $a = 1$ If a share of $a = 1$ is necessary, then $c > r_f$.

Other solutions are economically unfounded.



We have two equations for each case. First case rearranges to

$$(1 + r_f) \tilde{D}_t = \widetilde{CF}_{t+1}^1(d) + \tilde{D}_{t+1}(d) + a_{t+1} \tilde{V}_{t+1}^1(d) \quad (3)$$

with a solution $a \in (0, 1)$ furthermore $c = r_f$.

If, on the other hand, $a = 1$ gives us

$$(1 + r_f) \tilde{D}_t = (1 + c_t) \tilde{D}_t Q(u) + (\widetilde{CF}_{t+1}^1(d) + \tilde{V}_{t+1}^1(d)) Q(d) \quad (4)$$

for the coupon.



Coupon (c) and partial asset sale rate (a) are to be determined in insolvency:

$$\begin{array}{ll} \text{either } c = r_f & \implies \text{partial asset sale} \\ \text{or } a = 1 & \implies \text{complete asset sale.} \end{array}$$

Because of asset sale we distinguish between ex ante and ex post cash flows and values.

